BMI Calculator Project

Statement

Problem Statement

Create a Python program that acts as a simple BMI Calculator. The program should accept the user's name, age, gender, height in centimetres and weight in kilograms. It should then calculate BMI, display the result clearly, show the programmed BMI category and identify the user's age group.

Objectives

- To develop an interactive Python program using console input and output.
- To practise creating and calling user-defined functions.
- To apply mathematical calculations using Python operators.
- To use conditional statements for decision-making.
- To use a loop so that more than one calculation can be performed.
- To validate height and weight before calculation.
- To present the final result in a clear and organised format.

Input Requirements

The program requests five values from the user: name, age, gender, height in centimetres, and weight in kilograms. Age is converted to an integer, while height and weight are converted to floating-point numbers.

Processing

Height is first converted from centimetres to metres by dividing it by 100. BMI is then calculated using the formula weight ÷ (height × height). The resulting value is rounded to two decimal places.

The program next evaluates the BMI using the conditional ranges written in the code. It also evaluates age to place the user into one of four programmed age groups.

Output

The program displays the entered details, height in metres, weight, BMI, BMI category, a BMI range reference, age group and a short programmed suggestion. It then asks whether the user wants to perform another calculation.

Expected Behaviour

If the entered height or weight is zero or negative, the program does not perform the calculation and asks the user to enter valid values. If the user enters “yes” when asked whether to calculate again, the program starts another calculation. If the user enters “no”, the program displays a closing message and ends.

Educational Scope

This project is designed to demonstrate Python programming concepts through a practical example. The categories and messages are part of the program's educational logic and are not a substitute for professional health assessment.